List,Tuple,Dictionary

1. List

```
thisList = ["apple","banana","Cherry","dragonfruit"]
print(thisList)

thisList = list(("apple","banana","Cherry","dragonfruit"))
print(thisList)

['apple', 'banana', 'Cherry', 'dragonfruit']
```

List allows duplicate,it is changeable and ordered(like array)

```
print(type(thisList))

<class 'list'>
```

1. Len

```
print(len(thisList))

4

list1 = ["abc",34,True,40,'z']
print(list1)
print(type(list1))

['abc', 34, True, 40, 'z']
<class 'list'>
```

The slicing and element callling in list is similar to that of string

```
print("list1[2] = ",list1[2])

list1[2] =  True

print("list1[:2] = ",list1[:2])

list1[:2] =  ['abc', 34]

print("list1[-1] = ",list1[-1])

list1[-1] =  z

fruit = ["apple","banana","cherry","orange","Kiwi","Melon","Mango"]
```

```
print(fruit)
print(fruit[2:5])
print(fruit[:4])
```

```
['apple', 'banana', 'cherry', 'orange', 'Kiwi', 'Melon', 'Mango']
['cherry', 'orange', 'Kiwi']
['apple', 'banana', 'cherry', 'orange']
```

```
print(fruit[-4:-1])
```

```
['orange', 'Kiwi', 'Melon']
```

```
fruit[1]="blackcurrent"
```

```
print(fruit)
```

```
['apple', 'blackcurrent', 'cherry', 'orange', 'Kiwi', 'Melon',
'Mango']
```

```
fruit[2:4]=["chickoo","dragonfruit"]
```

```
print(fruit)
```

```
['apple', 'blackcurrent', 'chickoo', 'dragonfruit', 'Kiwi', 'Melon',
'Mango']
```

```
fruit.append("pineapple")
```

```
print(fruit)
```

```
['apple', 'blackcurrent', 'chickoo', 'dragonfruit', 'Kiwi', 'Melon',
'Mango', 'pineapple']
```

```
fruit.insert(3,"watermelon")
```

```
print(fruit)
```

```
['apple', 'blackcurrent', 'chickoo', 'watermelon', 'dragonfruit',
'Kiwi', 'Melon', 'Mango', 'pineapple']
```

```
tropical=["papaya","jackfruit","gauva"]
fruit.extend(tropical)
```

```
print(fruit)
```

```
['apple', 'blackcurrent', 'chickoo', 'watermelon', 'dragonfruit',
'Kiwi', 'Melon', 'Mango', 'pineapple', 'papaya', 'jackfruit', 'gauva']
```

```
fruittuple=("lichi","grapes")
```

```
fruit.extend(fruittuple)
```

```
print(fruit)
```

```
['apple', 'blackcurrent', 'chickoo', 'watermelon', 'dragonfruit',
'Kiwi', 'Melon', 'Mango', 'pineapple', 'papaya', 'jackfruit', 'gauva',
'lichi', 'grapes']
```

```
fruit.remove("papaya")
print(fruit)
```

```
['apple', 'blackcurrent', 'chickoo', 'watermelon', 'dragonfruit',
'Kiwi', 'Melon', 'Mango', 'pineapple', 'jackfruit', 'gauva', 'lichi',
'grapes']
```

```
fruit.pop(2)
print(fruit)
```

```
['apple', 'blackcurrent', 'watermelon', 'dragonfruit', 'Kiwi',
'Melon', 'Mango', 'pineapple', 'jackfruit', 'gauva', 'lichi',
'grapes']
```

```
del fruit[5]
print(fruit)
```

```
['apple', 'blackcurrent', 'watermelon', 'dragonfruit', 'Kiwi',
'Mango', 'pineapple', 'jackfruit', 'gauva', 'lichi', 'grapes']
```

```
#fruit.clear()
#print(fruit)
```

```
# del fruit
# print(fruit)
```

```
for x in fruit:
    print(x)
```

```
apple
blackcurrent
watermelon
dragonfruit
Kiwi
Mango
pineapple
jackfruit
gauva
lichi
grapes
```

```
print("apple" in fruit)
```

```
True
```

```
fruit.sort()
print(fruit)
```

```
['Kiwi', 'Mango', 'apple', 'blackcurrent', 'dragonfruit', 'gauva',
'grapes', 'jackfruit', 'lichi', 'pineapple', 'watermelon']
```

```
fruit.sort(reverse=True)
print(fruit)
```

```
['watermelon', 'pineapple', 'lichi', 'jackfruit', 'grapes', 'gauva',
'dragonfruit', 'blackcurrent', 'apple', 'Mango', 'Kiwi']
```

```
!jt -t onedork -T -N
```

```
fruit2 = fruit#updates both list even after altering
print(fruit2)
fruit.append("berry")
print(fruit)
print(fruit2)
```

```
['watermelon', 'pineapple', 'lichi', 'jackfruit', 'grapes', 'gauva',
'dragonfruit', 'blackcurrent', 'apple', 'Mango', 'Kiwi']
['watermelon', 'pineapple', 'lichi', 'jackfruit', 'grapes', 'gauva',
'dragonfruit', 'blackcurrent', 'apple', 'Mango', 'Kiwi', 'berry']
['watermelon', 'pineapple', 'lichi', 'jackfruit', 'grapes', 'gauva',
'dragonfruit', 'blackcurrent', 'apple', 'Mango', 'Kiwi', 'berry']
```

```
fruit2 = fruit.copy()
print(fruit2)#doesnt updates
fruit.append("blueberry")
print(fruit)
print(fruit2)
```

```
['watermelon', 'pineapple', 'lichi', 'jackfruit', 'grapes', 'gauva',
'dragonfruit', 'blackcurrent', 'apple', 'Mango', 'Kiwi', 'berry']
['watermelon', 'pineapple', 'lichi', 'jackfruit', 'grapes', 'gauva',
'dragonfruit', 'blackcurrent', 'apple', 'Mango', 'Kiwi', 'berry',
'blueberry']
['watermelon', 'pineapple', 'lichi', 'jackfruit', 'grapes', 'gauva',
'dragonfruit', 'blackcurrent', 'apple', 'Mango', 'Kiwi', 'berry']
```

```
fruit3 = fruit + fruit2
fruit3.sort()
print(fruit3)
```

```
['Kiwi', 'Kiwi', 'Mango', 'Mango', 'apple', 'apple', 'berry', 'berry',
'blackcurrent', 'blackcurrent', 'blueberry', 'dragonfruit',
'dragonfruit', 'gauva', 'gauva', 'grapes', 'grapes', 'jackfruit',
'jackfruit', 'lichi', 'lichi', 'pineapple', 'pineapple', 'watermelon',
'watermelon']
```

```
for x in fruit2:
    fruit.append(x)
```

```
fruit.sort()
print(fruit)
```

```
['Kiwi', 'Kiwi', 'Mango', 'Mango', 'apple', 'apple', 'berry', 'berry',
'blackcurrent', 'blackcurrent', 'blueberry', 'dragonfruit',
'dragonfruit', 'gauva', 'gauva', 'grapes', 'grapes', 'jackfruit',
'jackfruit', 'lichi', 'lichi', 'pineapple', 'pineapple', 'watermelon',
'watermelon']
```

```
berries = ["blueberry","strawberry","mulberry"]
berries*3
print(berries)
```

```
['blueberry', 'strawberry', 'mulberry']
```

```
sq = [x**2 for x in range(1,6)]
print(sq)
```

```
[1, 4, 9, 16, 25]
```

```
even = [x for x in range(10) if x%2==0]
print(even)
```

```
[0, 2, 4, 6, 8]
```

```
upper_fruit=[w.upper() for w in fruit]
print(upper_fruit)
```

```
['KIWI', 'KIWI', 'MANGO', 'MANGO', 'APPLE', 'APPLE', 'BERRY', 'BERRY',
'BLACKCURRENT', 'BLACKCURRENT', 'BLUEBERRY', 'DRAGONFRUIT',
'DRAGONFRUIT', 'GAUVA', 'GAUVA', 'GRAPES', 'GRAPES', 'JACKFRUIT',
'JACKFRUIT', 'LICHI', 'LICHI', 'PINEAPPLE', 'PINEAPPLE', 'WATERMELON',
'WATERMELON']
```

nested list

```
a = [[11,12,13],[21,22,23],[31,32,33]]
print(a)
```

```
[[11, 12, 13], [21, 22, 23], [31, 32, 33]]
```

Tuple

tuple is immutable

```
tfruit=("apple","banana","chickoo","dragonfruit","grape","gauva","jack
fruit","lemon")
print(tfruit)
print(type(tfruit))
```

```
('apple', 'banana', 'chickoo', 'dragonfruit', 'grape', 'gauva',
'jackfruit', 'lemon')
<class 'tuple'>
```

indexing in tuple is similar to list

```
print(type(tfruit[0]))
print(tfruit[0])

<class 'str'>
apple

print(len(tfruit))

8

print(tfruit[1:2])
print(tfruit[:-1])
print(tfruit[-2:])

('banana',)
('apple', 'banana', 'chickoo', 'dragonfruit', 'grape', 'gauva',
'jackfruit')
('jackfruit', 'lemon')
```

we can update a element in tuple by first convertin it into a list and then do changes and convert back to a tuple

```
x=tfruit
y=list(x)
y[1]="kiwi"
x=tuple(y)
print(x)

('apple', 'kiwi', 'chickoo', 'dragonfruit', 'grape', 'gauva',
'jackfruit', 'lemon')

t=('JP morgan','20','education')
print(t)

('JP morgan', '20', 'education')

(company,empid,profile)=t
print(company)
print(empid)
print(profile)

JP morgan
20
education
```

```
t1=(1,2,3,4)
a,b,c,d=t1
print(a,b,c,d)
```

```
1 2 3 4
```

```
a,b,c=t1
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
<ipython-input-52-408f20357701> in <module>
----> 1 a,b,c=t1

ValueError: too many values to unpack (expected 3)
```

```
(a,b,c)=t1[0:3]
```

```
print(a,b,c)
```

```
1 2 3
```